

ECON 2006 · MODULE 1 · CHAPTERS 1 & 2

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# Introduction to Economics Scarcity, Choice & Opportunity Cost

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# What is economics?

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*"Economics is the study of how people make choices when they can't have everything they want."*

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Resources are limited. Wants are not. Scarcity forces choice — and where there is choice, there are trade-offs, incentives, and consequences worth studying.

## TWO LENSES

# Micro zooms in. Macro zooms out.

### MICROECONOMICS

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*One consumer. One firm. One market.*

How prices, wages, and trade-offs are set when a single decision-maker faces a single choice.

→ Why Apple prices the iPhone at \$999

→ Why your roommate picked CS over biology

### MACROECONOMICS • THIS COURSE

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*The whole economy in one frame.*

Millions of households and firms, summed into national totals.

→ Why US GDP fell 8% in 2020

→ Why prices jumped 9% in 2022

## INTRODUCTION

# Why does it matter?

- Economics shapes **every aspect of daily life** — from government policy to personal finance.
- It answers the questions that shape the world:
  - Why do recessions happen?
  - What drives inflation or unemployment?
  - How do trade wars affect jobs?
- Making informed decisions — personal, professional, civic — requires **economic reasoning**.

## STAKES

# Three reasons this matters — *even if you're not majoring in econ.*

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**01**

### **Career**

Your industry lives inside the macro environment. Rates decide funding, inflation decides whether your raise is real, recessions decide whether you keep your job.

**02**

### **Citizenship**

Every election is partly a referendum on macro policy — taxes, deficits, the Fed, trade. Better to vote informed.

**03**

### **Your Life**

Mortgages, savings, jobs — you will make these decisions whether you understand them or not. Understanding is cheaper than not.



# Scarcity & Choice

Why do we need economics at all? Because resources are limited — and choices must be made.

# The Central Problem

**Resources are scarce — so every society must make choices.**

All of economics tries to answer three fundamental questions:

**01**

**What to produce?**

Which goods and services to make?

**02**

**How to produce it?**

Which inputs and methods to use?

**03**

**For whom?**

How is output distributed?

## TRADE-OFFS

# Every choice has a cost.

*"There ain't no such thing as a free lunch."*

— Milton Friedman, 1975

### THE STUDENT

Tonight: study, sleep, or go out. You can't do all three.  
Whichever you skip is the cost of what you chose.

### THE FARMER

Plant corn or soybeans? The same acre can't grow both.  
Soybeans forgone is the cost of corn planted.



# Opportunity Cost

A formal way to measure the true cost of any choice — the best thing you give up.

## OPPORTUNITY COST

# Definition

The value of the **best alternative forgone** when a choice is made.

### EXAMPLE — \$10,000 DECISION

You have \$10,000. Options: pay tuition · buy a car · invest in stocks · take a vacation.

**If you choose tuition,** the opportunity cost is the value of your single best remaining option — whichever of the others you valued most.

Opportunity cost is **not** the sum of all forgone options — only the **best one**.



# Production Possibility Frontier

Visualizing scarcity, trade-offs, and opportunity cost  
for an entire economy.

## PRODUCTION POSSIBILITY FRONTIER

# Setting Up the Model

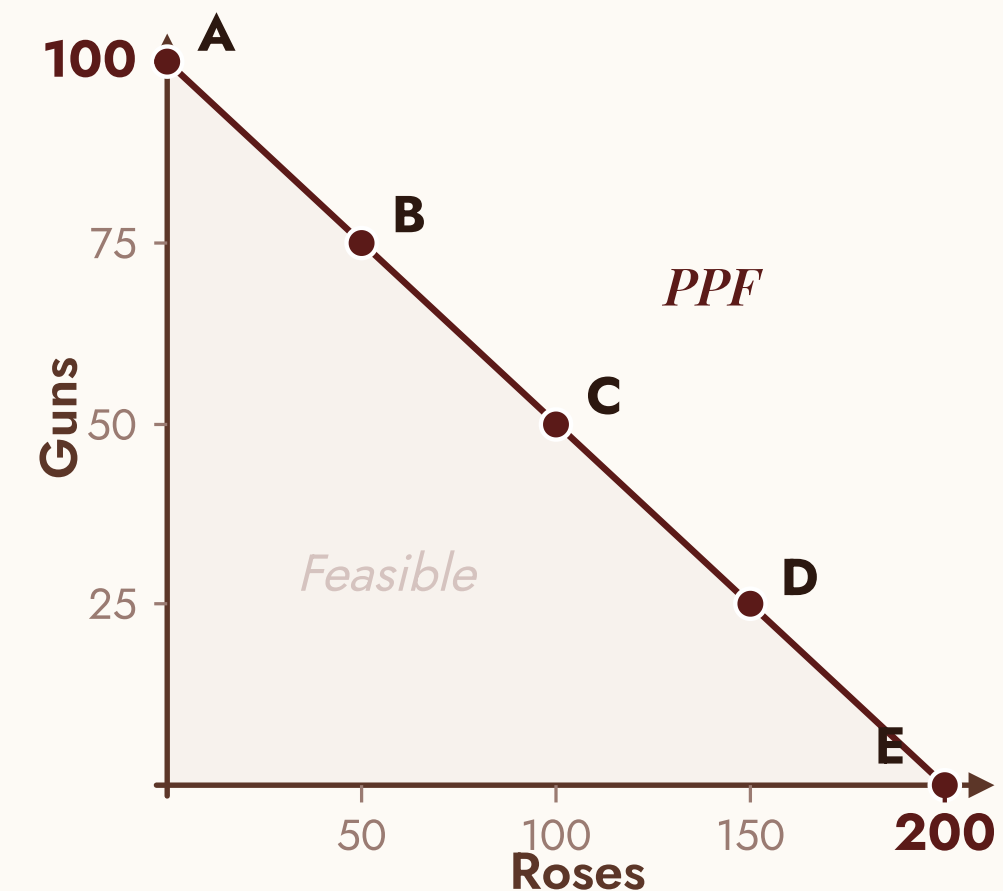
A simple economy — **100 acres of land**, two goods:

- **Guns** — 1 acre each → max 100
- **Roses** — ½ acre each → max 200

### PPF — DEFINITION

All combinations of goods producible when **all resources are used efficiently**.

Constant opportunity cost → straight-line PPF

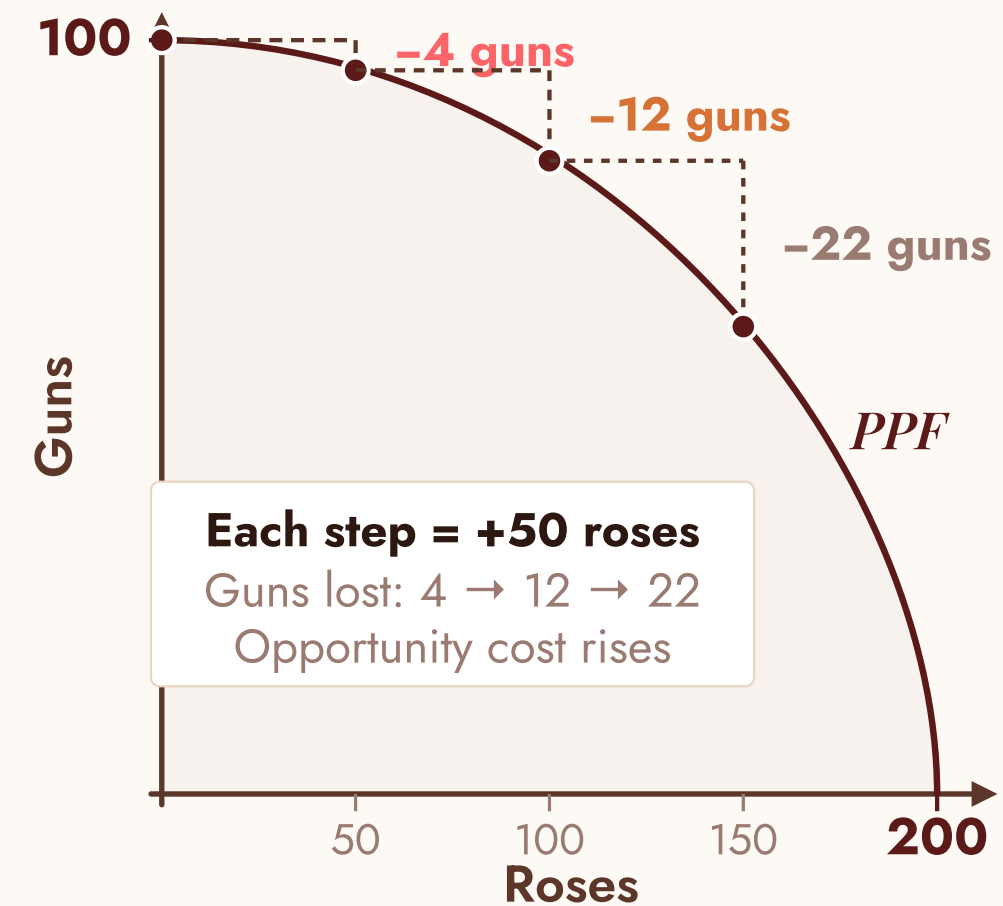


## PRODUCTION POSSIBILITY FRONTIER

# Is the PPF Always a Straight Line?

In practice: **No**. Resources are not perfectly interchangeable — shifting more land to roses makes each additional rose progressively costlier.

- **Law of Increasing Opportunity Costs** — each extra unit costs more of the other good
- PPF bows **outward** from the origin
- Slope = **Marginal Rate of Transformation (MRT)**



## PRODUCTION POSSIBILITY FRONTIER

# Where Can We Operate?

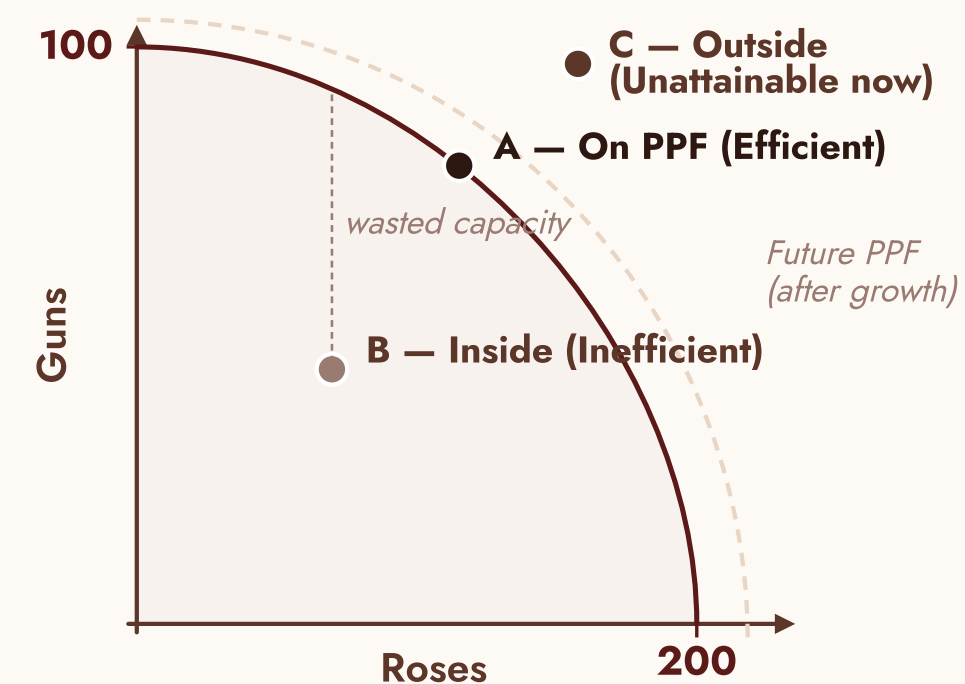
- **On the PPF** — all resources used efficiently
- **Inside the PPF** — resources are idle or technology underused

Why might we be inside?

- Unemployment — workers without jobs
- Technology not used to its full potential

### SHIFTING OUTWARD = ECONOMIC GROWTH

More resources or better technology expands the PPF — the economy can produce more of everything.



## SCARCITY & CHOICE

# How Are Resources Allocated?

Given scarcity, which mechanism decides what gets produced and who gets it?

### Command Economy

Government decides what, how, and for whom to produce.  
Central planning replaces the market.

*Former USSR · Cuba · North Korea*

### Market Economy

**Price** is the key signal — it directs resources to their highest-valued uses through supply and demand.

*Farmer's crop choice · Your choice of major*

### COMING UP NEXT

Since price is central to the market mechanism, we need to understand how it's determined. That's where Demand & Supply comes in.

## WRAP-UP

# What We Covered

- 1 Economics studies how **scarce resources** are allocated — micro (individual markets) and macro (economy-wide).
- 2 Scarcity forces choices. Every choice has a **trade-off** — a cost and a benefit that must be weighed.
- 3 **Opportunity cost** = value of the best alternative forgone — the formal price of any decision.
- 4 The **PPF** maps all efficient combinations; its concave shape reflects the Law of Increasing Opportunity Costs.

LOOKING AHEAD

# Up Next: Demand & Supply

*We've seen why choices must be made. Now: how does a market economy make them?*

## **Price is the answer**

Price signals guide what gets produced, how much, and who gets it. Topic 2 explores how prices emerge through Demand and Supply.

Read ahead: Chapters 3 & 4 — Demand, Supply, and Market Equilibrium